Project 1, Milestone 3

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Credit Card Fraud: How it impacts consumers, trends in the current landscape,

and what everyone can do to prevent it

Business Problem

Credit card fraud is when an unauthorized person uses a card to make unauthorized purchases or withdrawals by using physical card theft, card skimming devices or online information stealing to receive the card information (OCC, 2025).

There are two sides to the problem of credit card fraud. The first being the impact of credit card fraud on consumers. The second being the trends in the credit card fraud landscape and what businesses can do to mitigate it as much as possible. Credit card fraud can be prevented by both the consumer and business by using safe credit card handling methods and having adequate prevention strategies in place. Regardless, consumers need to know what they can do when it impacts them.

Background

Identity theft comes in many different forms but credit card fraud is the most common during the first three quarters of 2025. There were 503,450 reported cases which was a 54% growth over prior year. New account [an identity thief uses a person's information to open a new card in that person's name] and existing account [a thief uses your existing account information to make fraudulent purchases] are the two types of credit card fraud and new account fraud accounts for 90% of all credit card fraud (Caporal, 2026).

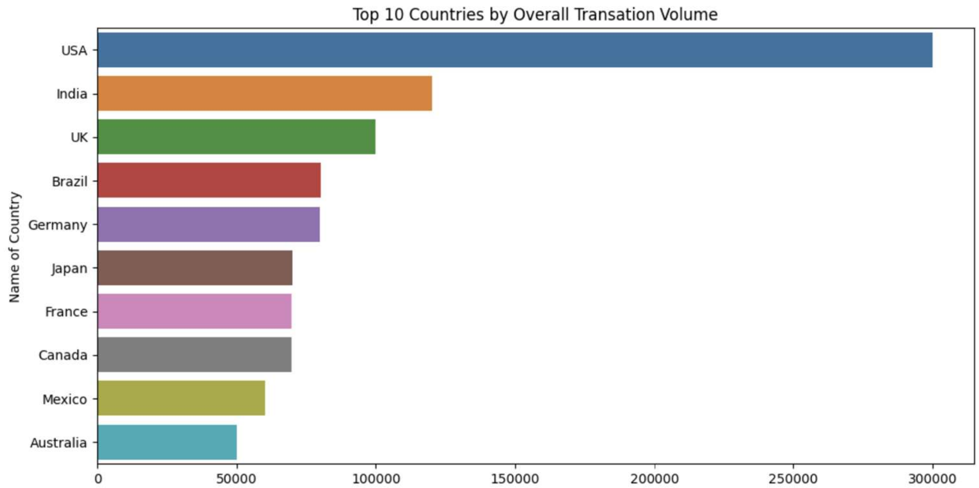
For those people that were found and sentenced for credit card fraud, 74.7% were men, 55.5% had little to no criminal history, 82.8% were United States citizens with an average age of 38 years old, the median loss was $154,919, and were sentenced to an average of 26 months in prison. The average sentence imposed has remained steady from 2020 to 2024 (USSC, 2024).

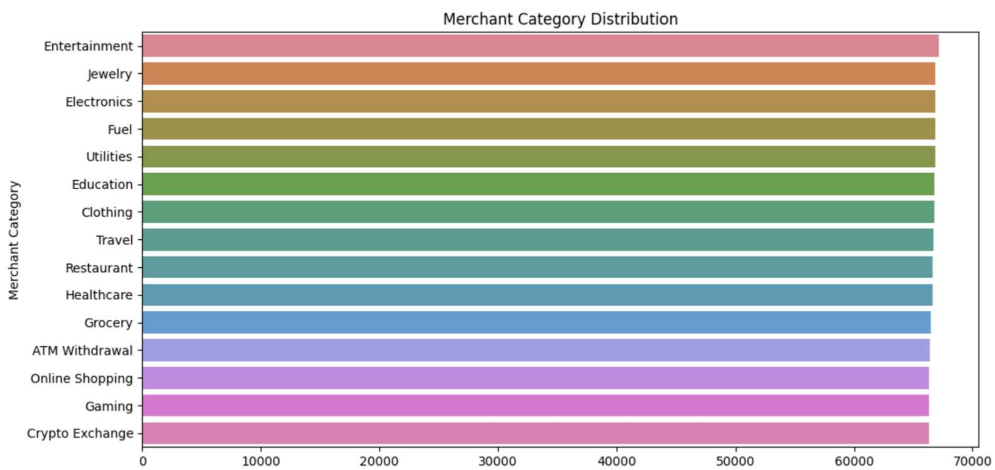
Data Explanation

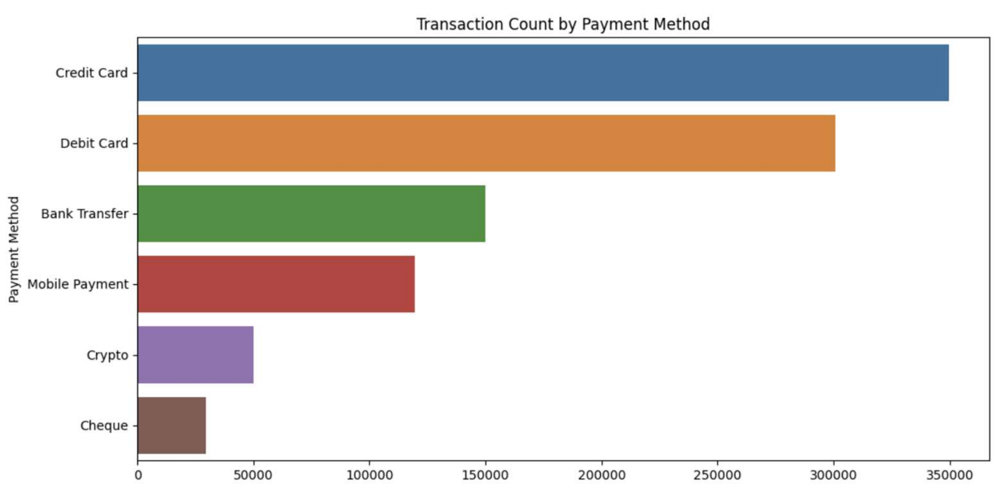
Through discovery, two potential data sets were found both through Kaggle. The first is a synthetic data set with 1,000,000 records with 26 columns of data related to the transaction as well as the customer and account (Islam, 2026). The second was a data set generated by Vesta's real world transaction data and created as a competition data set within Kaggle. There were 871 columns with no data dictionary defined and column names with a simple letter and number combinations (IEEE Computational Intelligence Society, 2019). While the data was already provided in test (506,691 rows) and training (590,541 rows) data sets the ambiguity of the column names made interpretation of the potential results more difficult. Having worked in the financial industry, the synthetic data seemed to have the most relevant data elements that a bank would use in their fraud mitigation strategies.

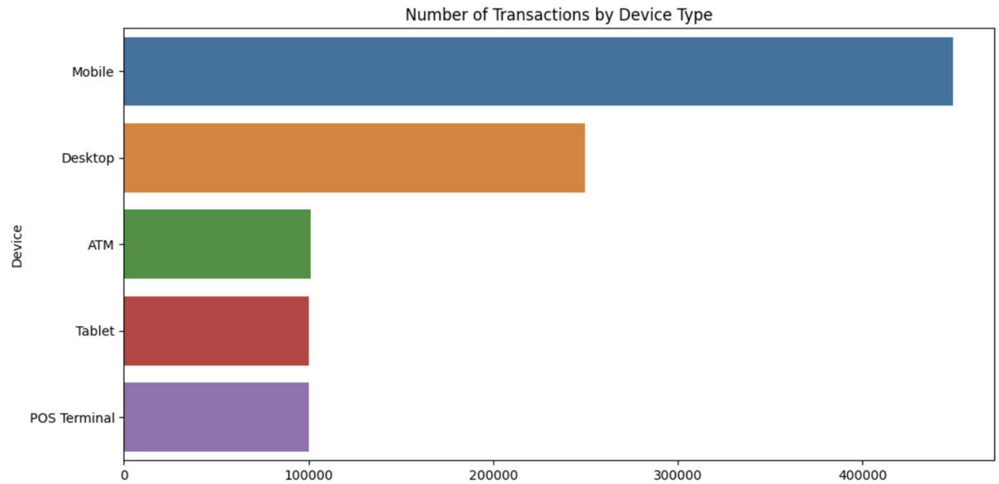
The synthetic data came from the Bank Transaction Fraud Detection Dataset (BTFDD). It has 26 total columns with data about the transaction(time, date, hour of day, weekend indicator, night indicator, country, city, merchant category, payment method, device type, distance from home (km), time since last transaction (hours), international indicator, failed attempts count, transaction amount, number of previous transactions, monthly transaction frequency) and the customer (age, credit score, age of account, account balance, pin changed recently) as well as an indicator to if the transaction was fraudulent and the type of fraud (Islam, 2026).

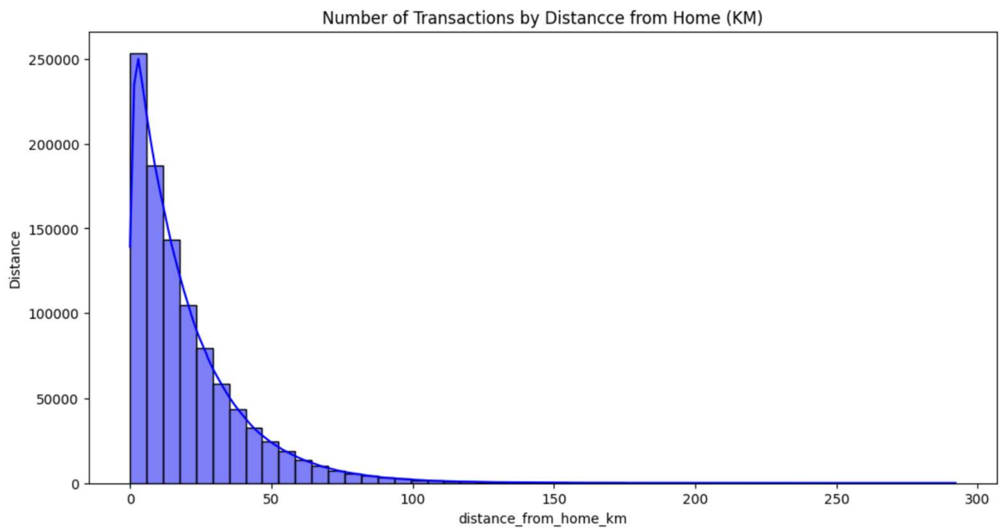
Exploratory Data Analysis

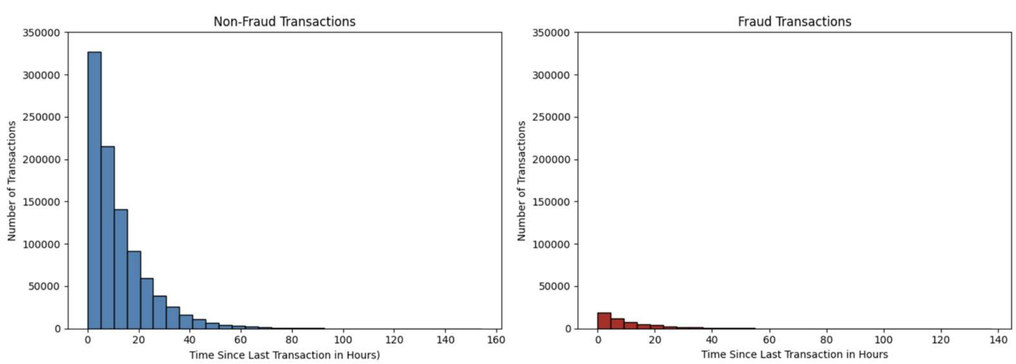


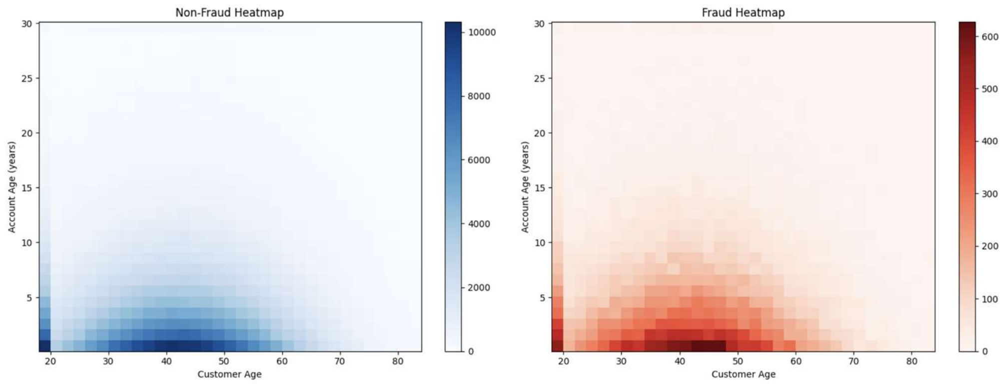





Number of Transactions by Device Type
Mobile
Desktop
ATM
Tablet
POS Terminal
Device
0
100000
200000
300000
400000


Number of Transactions by Distancce from Home (KM)
Distance
distance_from_home_km
0
50000
100000
150000
200000
250000
0
50
100
150
200
250
300


Non-Fraud Transactions
Number of Transactions
Time Since Last Transaction in Hours)
Fraud Transactions
Number of Transactions
Time Since Last Transaction in Hours



Methods

The file provided through the BTFDD data set was very clean, had no nulls in any column, and was appropriately formatted. Before modeling I removed columns that the model cannot evaluate such as ID, date/time, country, and city. I then one hot encoded the categorical fields. Split the data into test and train groups with the test size of 20% of the rows. A logistic regression model was ran with a balanced class weight, the liblinear solver, a standard scaler, with max iterations of 5,000.

Analysis

The model was created with 98.42% accuracy. The precision the model indicates that about 86% of transactions flagged are actual fraud. The recall shows the model catches 83% of all fraud cases. The false positives are 1,374 and false negatives are 1,795.

Conclusion

False positives have more of an impact on the business as the customer as a non-fraudulent transaction causes the customer must either find an alternative form of payment or contact the bank. False negatives show that some fraud will not be immediately identified. Customers will still need to work with their bank to dispute the charge(s), but this is the normal course of business for the company.

Assumptions

Overarching assumptions are that the people executing the fraud maintain their current strategies and do not adapt to how businesses are approving or declining transactions. Currently financial institutions are using pattern recognition, behavioral analysis, geo-location, two-factor authentication, network analysis, real-time monitoring, data analytics possibly through machine learning and artificial intelligence, employee training, and collaboration with other financial institutions (*Banking Fraud Detection*, 2024).

Limitations

As the dataset used is composed of synthetic data it may come to be that it is not relevant to the real world. If other data sets were used, they generally are from the late 2010's and the fraud environment fluctuates frequently making the findings no longer relevant.

Challenges

Not having access to the full bank data to identify columns to include in the model may limit its performance. Additionally the darknet is selling credit card information to fraudsters providing an unlimited supply of card information to perpetuate credit card fraud. So even if consumers are very careful with their information, data breaches occur and their information can be used for fraud (Howell & Maimon, 2022).

Future Uses

The credit card fraud environment is consistently changing so the model will need to be rerun frequently with current data to ensure that it maintains or improves its accuracy in identifying fraudulent transactions. Variables included need to be constantly evaluated to ensure the model has all possible options to include in the model.

Recommendations

"FinCEN data shows that credit and debit card fraud has steadily intensified over the past five years, evolving from a relatively modest baseline of 15k–20k monthly reports in 2020–2021 to a sustained surge above 30k per month by 2022. The elevated activity persisted through 2023, when filings hit a record 38k in March, and has since stabilized at a consistently high level of 30k–36k per month through mid-2025, underscoring that card-based fraud has become an entrenched and durable threat vector for U.S. financial institutions." (Maimon, 2025)

These statistics show that all financial institutions need a fraud mitigation strategy. Depending on their operational system, they should implement the most accurate model they can.

Implementation Plan

Financial institutions should implement some kind of fraud mitigation strategy. Models can be created for implementation into their operational system. Depending on the age of the operational system they may need to find a way to integrate a model into the transaction authorization process.

Also, consumers need to be educated on what to do if they have unauthorized transactions, which is to contact their bank immediately. They may have protection under federal laws. Use of a debit card requires notification within two days or they can be held responsible for up to $500 in unauthorized transactions (FDIC, 2026).

Ethical Assessment

Ethical considerations need to be made for geographic areas that come up during analysis that currently have heavy political issues or wars going on. But considerations should be made further than that. The primary concern is bias and fairness in automated decision making through use of AI models. This model removes geographic fields before training, but other fields may act like proxy geographic attributes inadvertently. These fields are device type, merchant category, distance from home, and/or transaction timing. Prior research has shown that fraud models can unintentionally discriminate when proxy variables correlate with protected classes (Barocas & Selbst, 2016). This raises the risk that certain consumer groups could experience disproportionately high false positive rates, leading to declined legitimate transactions or increased friction in accessing financial services.

As transactions are evaluated to be approved or declined through the lens of fraud, the false positives that occur can cause consumer harm and consequences of misclassification. Regulatory bodies such as the CFPB highlight that automated financial decisions must minimize consumer harm and avoid unfair or deceptive outcomes (Consumer Financial Protection Bureau, 2022). Institutions have an ethical responsibility to calibrate false positive and false negative thresholds carefully, provide rapid dispute resolution processes, and ensure that automated decisions do not unfairly penalize consumers.

The central ethical concern is data privacy and security. There is an ongoing supply chain through the darknet markets. This highlights the ethical duty of institutions to safeguard customer data, minimize unnecessary data retention, and invest in robust cybersecurity measures. Ethical stewardship requires institutions to protect consumer information not only to prevent fraud but also to avoid contributing to the broader ecosystem of data mistreatment. The FTC stresses that organizations must implement reasonable data security practices to protect consumers from preventable harm (Federal Trade Commission, 2023).

These considerations demonstrate that ethical fraud detection requires more than technical accuracy. It demands fairness, transparency, consumer protection, responsible data use, and continuous monitoring to ensure that fraud mitigation strategies serve both institutional needs and the well-being of the consumers they are designed to protect.

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